

2025 MOCK HSC EXAM

Mathematics Advanced

General**Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:**100****Section I – 10 marks** (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–27)

- Attempt Questions 11–31
- Allow about 2 hours 45 minutes for this section

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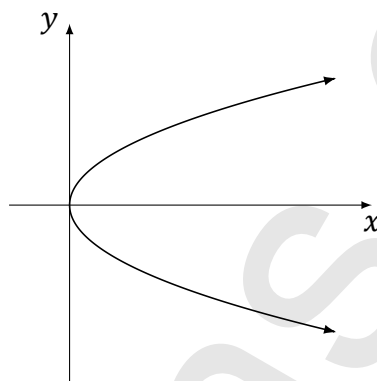
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10 marks

Allow about 15 minutes for this section

1 Consider the following relationship between x and y :



A. one-to-one

B. many-to-one

C. one-to-many

D. many-to-many

 $\log_2 x, \log_2 2, \log_2(5-x)$

A. 2

B. 3

C. 4

D. 5

3 What is the value of $\int_1^5 |x - 2| dx$?

A. 2

B. 3

C. 4

D. 5

- 4 The range of a function $f(x)$ which has domain all real x is $[-2, \infty)$.

What is the range of the function $g(x) = 3f(-2x) - 4$?

- A. $(-\infty, 4]$ B. $[-10, \infty)$ C. $[-2, \infty)$ D. $(-\infty, 6]$
- 5 Let C and D be two events with $P(C) = 0.2$, $P(D) = 0.5$, and $P(C|D) = 0.3$. What is the value of $P(D|C)$?
- A. 0.75 B. 0.6 C. 0.5 D. 0.3
- 6 The hair colour and eye colour of a class of 30 students were recorded. Students were grouped by the categories brown eyes and black hair. This is shown in the table below.

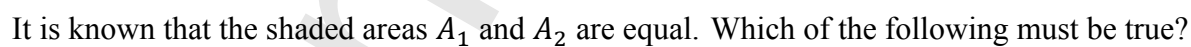
	Brown Eyes	Not Brown Eyes
Black Hair	12	2
Not Black Hair	10	6

A person from the class is chosen at random. Given that the person chosen has black hair, what is the probability that DO NOT have brown eyes?

- A. $\frac{1}{4}$ B. $\frac{1}{6}$
C. $\frac{1}{7}$ D. $\frac{1}{15}$
- 7 Which of the following is an odd function?
- A. $\frac{1}{x} \sin\left(\frac{1}{x}\right)$ B. $x^4 + x^2 + 1$
C. $x^5 + x^3 + x \ln x$ D. $\frac{1}{2} - \frac{1}{2^x + 1}$
- 8 If $f'(x) = \frac{4x}{4 + x^2}$ and $f(2) = f'(2)$, the value of $f(0)$ is:
- A. $2 \ln 2 + 1$ B. $4 \ln 2 - 1$ C. $1 - 2 \ln 2$ D. $1 - 4 \ln 2$

- Which of the following are possible values of h and k ?

- 10** Youssef sketches the following curve and three vertical lines, where $1 < \alpha < k < \beta$.



- 4 -

Section II

90 marks

Attempt Questions 11–31

Allow about 2 hours 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

Evaluate $\int x\sqrt{x^2 + 4} \, dx$.

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Question 12 (3 marks)

If $f(x) = \frac{1}{e^x \sin x}$, find the equation of the tangent at $x = \frac{\pi}{2}$.

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Question 13 (2 marks)

Differentiate $y = \ln\left(\frac{2x+1}{x^2+1}\right)$.

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Question 14 (3 marks)

Sketch the graph of $y = \frac{2x+1}{x-2}$, showing all important features.

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Question 15 (6 marks)

Let $f(x) = e^{2x} - 4e^x$.

- (a) Find the x -intercepts of $y = f(x)$.

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- (b) Find all the turning points of $f(x)$, and determine their nature.

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You may use the fact that $f'(x) = 2e^{2x} - 4e^x$ and $f''(x) = 4e^{2x} - 4e^x$ without proof.

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- (c) Sketch $y = f(x)$, showing all important features.

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Question 16 (3 marks)

Let $y = \frac{\ln x}{x}$.

(a) Find $\frac{dy}{dx}$.

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(b) Hence, evaluate $\int \frac{\ln x}{x^2} dx$.

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Question 17 (4 marks)

Let $f(x) = \frac{\sec x - \cos x}{\sec x + 1}$.

- (a) Show that $\frac{\sec x - \cos x}{\sec x + 1} = 1 - \cos x$.

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- (b) Hence, graph $y = f(x)$ for $0 \leq x \leq 2\pi$.

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Question 18 (5 marks)

Bob has won a small lottery and has been given two withdrawal options.

Option 1: Receive a lump sum of \$1 000 000.

Option 2: Receive a lump sum of \$500 000, and then receive annual payments of \$60 000 at the end of the year (including the first year) for ten years.

Bob will immediately deposit any payments into a bank account that earns 1.2% interest compounded monthly.

- (a) What is the value of Bob's earnings after ten years under Option 1? 1

- (b) What is the difference in the future earnings after ten years using Option 1 or Option 2? **3**

[illegible]

(c) What is the *present value* of Option 2?

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Question 19 (4 marks)

NSW Mathematics is reviewing the punctuality of their students. On any given day, there is a 60% chance that the weather condition is ‘good’. If the weather is ‘good’ then the probability that any given student arrives on time is 90%. If the weather is not ‘good’ then this probability drops to 60%.

Let G and A denote the events where the weather is ‘good’ and a given student arrives on time respectively.

(a) Draw a tree diagram to represent this situation.

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(b) Show that the probability that a given student arrives on time is 0.78.

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(c) Given that a randomly chosen student arrived on time, determine the probability that the weather was good.

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Question 20 (3 marks)

Let $f(x) = a \sin x + b \cos x$, where a and b are constants. If $f\left(\frac{\pi}{4}\right) = 3$ and $f'\left(\frac{\pi}{4}\right) = 1$, solve the equation $f(x) = 0$, for $0 \leq x \leq 2\pi$, correct to two significant figures. **3**

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Question 21 (3 marks)

The function $f(x) = e^{x^2}$ is continuous on the interval $[-1, 1]$. Use the trapezoidal rule with four subintervals to estimate the value of $\int_{-1}^1 e^{x^2} dx$. **3**

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Question 22 (4 marks)

oneeightseven records the number of hours spent training per week and the resting heart rate (in beats per minute) of six participants.

Training Hours (T)	2	3	4	6	7	10
Resting Heart Rate (H)	85	82	80	76	74	70

- (a) Find the equation of the least-squares regression line, rounding coefficients to the nearest integer, and use the equation to estimate the resting heart rate of someone who trains for 5 hours a week. 3

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- (b) oneeightseven trains for 25 hours a week to prepare for a marathon. Explain why the regression equation from part (a) may not be appropriate for estimating oneeightseven's resting heart rate. 1

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Question 23 (5 marks)

Consider the two curves $y = \ln x + 8x$ and $y = \ln x + x^2 + 7$.

- (a) Find the x value(s) at which the two curves intersect.

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- (b) Hence, find the area bound between the two curves.

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Question 24 (5 marks)

A discrete random variable X has the following probability distribution table

x	0	1	2	3	4
$P(X = x)$	a	$2a$	b	0.2	c

where a, b and c are constants.

It is known that $E(X) = 2$.

- (a) By finding expressions for b and c in terms of a , show that $0.05 \leq a \leq 0.18$.

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- (b) Hence deduce the maximum possible value for the variance of X .

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Question 25 (7 marks)

Abdullah wishes to purchase an item which she can resell for one dollar. Benjamin is selling this item and believes that the item is worth X dollars where X is a random variable with probability density

$$f(x) = \begin{cases} \alpha + \beta(2x - 1) & : 0 \leq x \leq 1 \\ 0 & : \text{otherwise} \end{cases}$$

where α and β are constants.

- (a) Show that $\alpha = 1$ and find the possible values of β for which $f(x)$ is a valid probability density function. **2**

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- (b) Abdullah places a bid b for the item and will *gain* the item only if her bid is *above* Benjamin's valuation of the item. Otherwise she will *lose* her bid. **1**

Show that her expected profit, E , is given by

$$E = F(b) - b$$

where $F(x)$ is the cumulative density function (cdf) of X .

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- (c) Suppose $\beta > 0$. Explain why Abdullah should either bid $b = 0$ or $b = 1$.

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- (d) Suppose $\beta < 0$. Find the value of b that maximises Abdullah's expected profit and find the expected profit in terms of β .

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You do NOT need to show that this is a maximum.

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Question 26 (4 marks)

Show that

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$$2 + 22 + 222 + 2222 + \cdots + \underbrace{2222 \dots 2}_{n \text{ 2's}} = \frac{2}{81}(10^{n+1} - 9n - 10).$$

maths.com

Question 27 (4 marks)

Let k be a real number. For what values of k does the equation $kx^2 + 3(k - 2)x = 9k - 9$ have two real roots for x ? 4

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Question 29 (5 marks)

Consider a random variable X that is normally distributed with mean 15 and standard deviation σ . It is known that $P(X < 10) = a$ and $P(X > 25) = b$.

- (a) Find $P(20 < X < 25)$ in terms of a and b . **2**

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- (b) Consider a random variable Y that is normally distributed with mean 25 and standard deviation 3σ . Find two pairs (m, n) such that $P(m < Y < n) = 1 - a - b$. **3**

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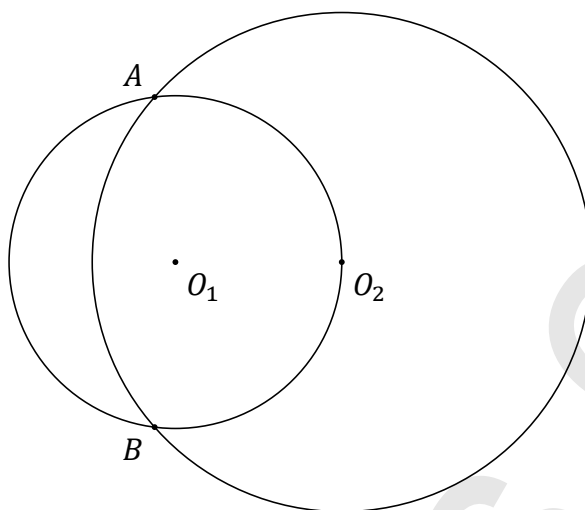
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Question 30 (5 marks)

In the figure below, the graphs of $C_1 : x^2 + y^2 = 4$ with centre O_1 and $C_2 : (x - 2)^2 + y^2 = 9$ with centre O_2 are shown without the coordinate axes. Their intersections are labelled A and B .



- (a) Find the value of $\cos \angle AO_1O_2$.

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Question 30 continues on page 24

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Question 31 (9 marks)

In the year 2050, NSWMaths manufactures and sells Ruiace sip stickers. Suppose

$$C(q) = \frac{7q^3 - 1000q^2 + 126000q}{100000} + 84$$

is the cost in millions of dollars to produce and sell q million units (you may assume this is the only cost). It is also known that the price each sticker can be sold for can be modelled by $D(q) = b - aq$, where a and b are positive constants. Define $P(q)$ to be the profit in millions of dollars associated with selling q million stickers.

- (a) From historical data, when 200 million units were produced, each unit sold for \$1.50, and when 50 million units were produced and sold, a profit of \$44.25 million was made.

- (i) Explain why $b = 1.5 + 200a$.

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- (ii) Write down an expression for $P(q)$ in terms of a , b , and q .

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Question 31 continues on page 26

(iii) Hence, show that

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$$P(q) = q \left(\frac{25}{6} - \frac{q}{75} \right) - \left(\frac{7q^3 - 1000q^2 + 126000q}{100000} + 84 \right).$$

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Question 31 continues on page 27

[illegible]

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*I want to learn more and more to see as beautiful what is necessary in things;
then I shall be one of those who make things beautiful.*

— Friedrich Nietzsche, Die fröhliche Wissenschaft.

EXAMINERS

one eight seven	Q2, 5, 20, 21, 22, 29
MokSifu	Q1, 6, 7, 9, 11, 13, 14, 17, 18, 24, 28, 30
Hugh Entwistle	Q18, 19, 24, 25
MagiicMushroom	Everything else.

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

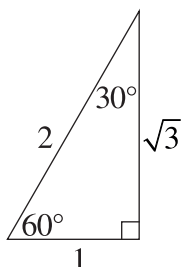
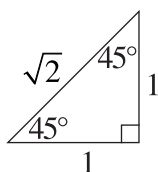
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

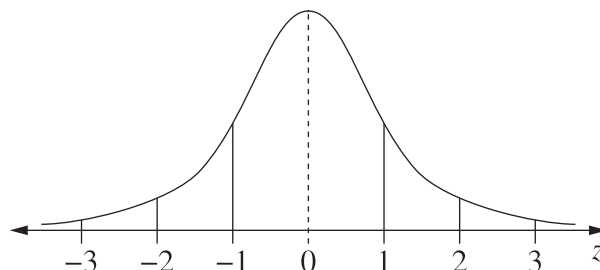
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$